

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

**Week 5: Generative Adversarial Networks (GANs):
Foundations**

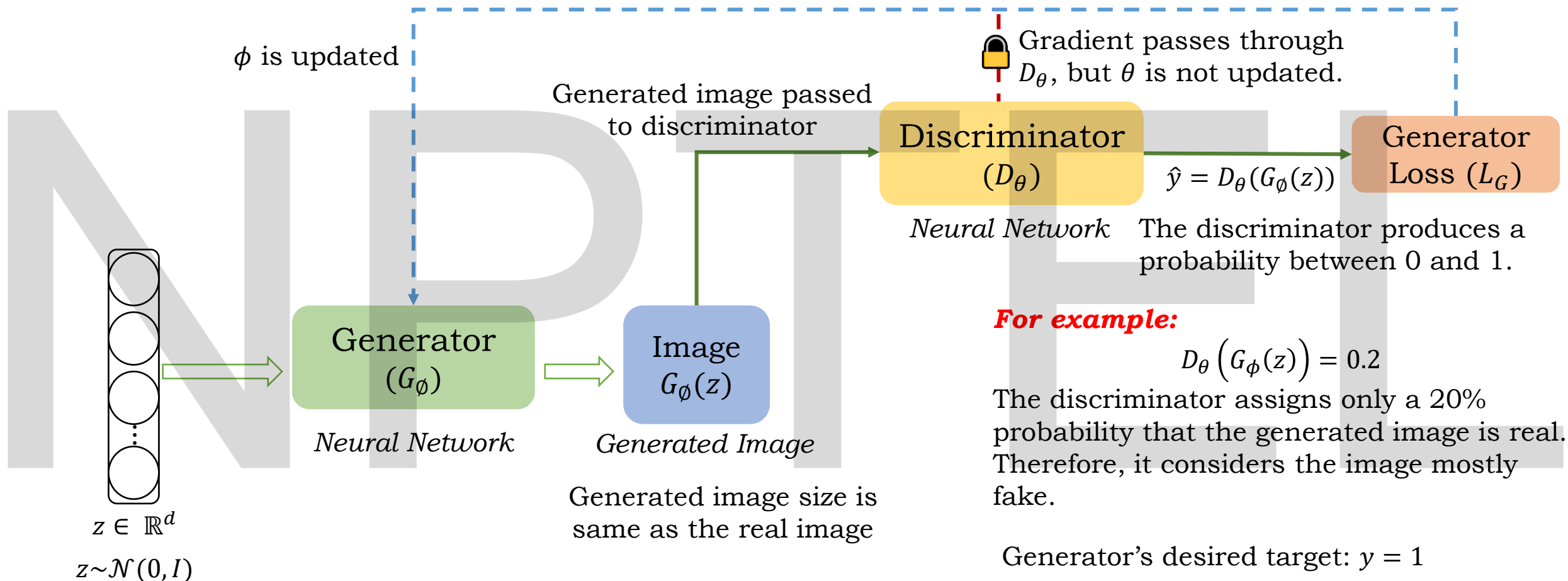
Today's Session Overview

GAN Architecture

- Generator
- Discriminator

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Generator Training



Generator Loss

Consider BCE Loss:

$$L_{BCE} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

During generator training, the generator wants the discriminator's prediction for the generated image to become close to 1.

Generator's goal: make the discriminator classify the generated image as real.

So,

$$y = \text{desired target} = 1$$

Since $y = 1$,

$$\begin{aligned} L_G &= -[1 \cdot \log \hat{y} + (1 - 1) \log(1 - \hat{y})] \\ &= -\log \hat{y} \end{aligned}$$

Since

$$\begin{aligned} \hat{y} &= D_\theta(G_\phi(z)) \\ L_G &= -\log D_\theta(G_\phi(z)) \end{aligned}$$

Forward pass:

$$z \rightarrow G_\phi(z) \rightarrow D_\theta \rightarrow \hat{y} = D_\theta(G_\phi(z)) \rightarrow L_G$$

Suppose:

$$\hat{y} = \underline{0.2},$$

Then:

$$L_G = -\log(0.2) \approx \underline{1.609} \checkmark \uparrow$$

The generator loss is high because the discriminator $\checkmark$ confidently identifies the generated image as fake.

Backward Pass: Generator

The gradient flows in the reverse direction:

$$L_G \rightarrow \hat{y} \rightarrow \hat{x} \rightarrow \phi$$

$$L_G \rightarrow D_\theta \rightarrow G_\phi \rightarrow \phi$$

Since:

$$\hat{y} = D_\theta \left(G_\phi(z) \right)$$

and

$$\hat{x} = G_\phi(z),$$

Using the chain rule:

$$\frac{\partial L_G}{\partial \phi} = \frac{\partial L_G}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial \hat{x}} \cdot \frac{\partial \hat{x}}{\partial \phi}$$

Substituting the definitions:

$$\frac{\partial L_G}{\partial \phi} = \frac{\partial L_G}{\partial D_\theta \left(G_\phi(z) \right)} \cdot \frac{\partial D_\theta \left(G_\phi(z) \right)}{\partial G_\phi(z)} \cdot \frac{\partial G_\phi(z)}{\partial \phi}$$

Then the generator parameters are updated using gradient descent:

$$\phi \leftarrow \phi - \eta \frac{\partial L_G}{\partial \phi}$$

where η is the learning rate.

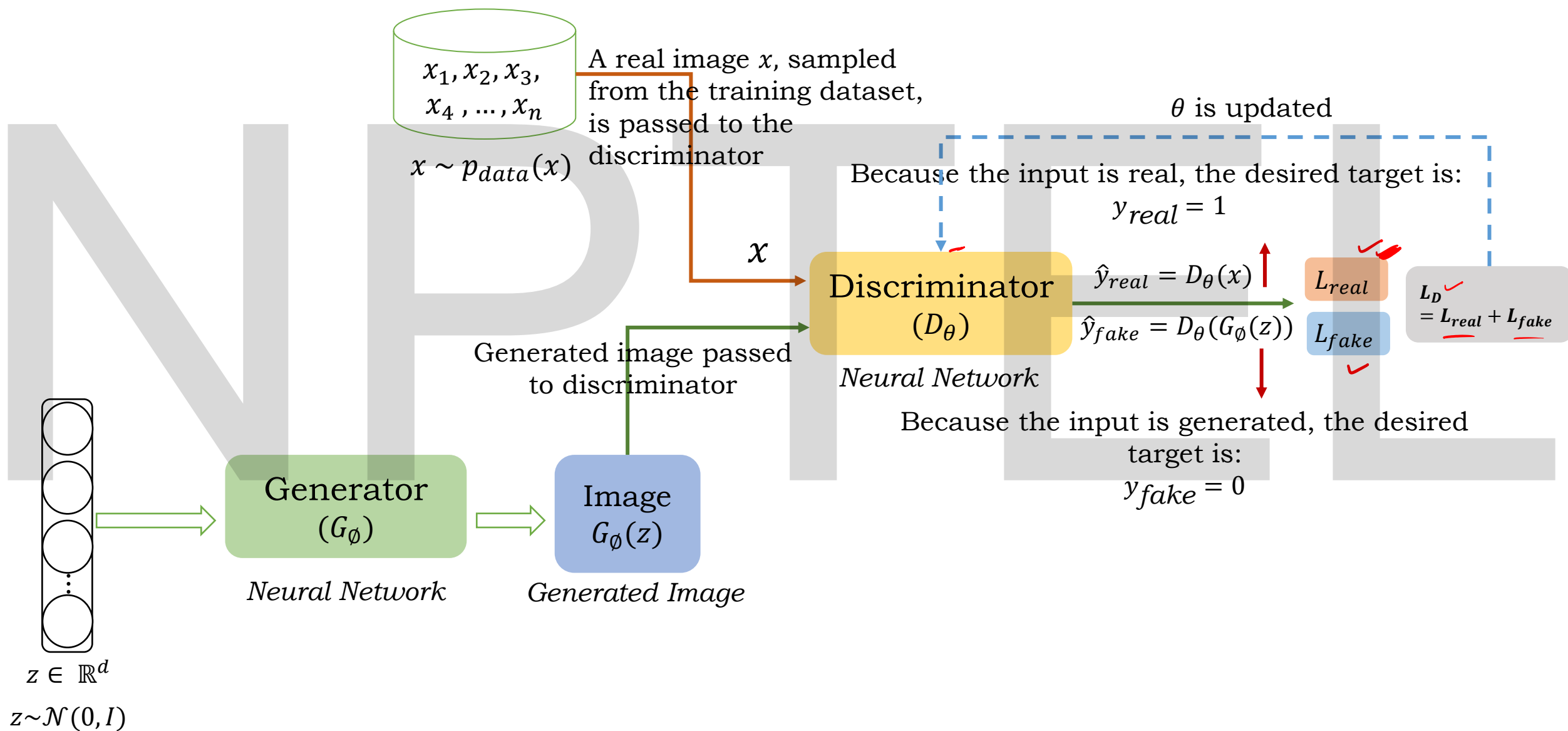
During this step: ϕ is updated and θ is frozen

The purpose of the update is to modify the generator so that the discriminator assigns a higher probability of being real to future generated images:

$$\underline{D_\theta \left(G_\phi(z) \right)} \rightarrow 1 \quad \checkmark$$

The discriminator is used to compute the gradient, but its parameters remain fixed during the generator update.

Discriminator Training



Discriminator Loss

Consider BCE Loss:

$$L_{BCE} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

During discriminator training, there are two cases:

Real image: $y = y_{real} = 1$

1. Loss for the real image

For a real image x :

$$\hat{y}_{real} = D_{\theta}(x)$$

Since:

$$y_{real} = 1$$

the BCE loss becomes:

$$L_{real} = -[1 \cdot \log \hat{y}_{real} + (1 - 1) \log(1 - \hat{y}_{real})] = -\log \hat{y}_{real}$$

Therefore:

$$L_{real} = -\log D_{\theta}(x)$$

The discriminator wants:

$$D_{\theta}(x) \rightarrow 1$$

2. Loss for the generated image

Generated image: $y = y_{fake} = 0$

For a generated image:

$$G_{\phi}(z)$$

and:

$$\hat{y}_{fake} = D_{\theta}(G_{\phi}(z))$$

Since:

$$y_{fake} = 0$$

the BCE loss becomes:

$$\begin{aligned} L_{fake} &= -[0 \cdot \log \hat{y}_{fake} + (1 - 0) \log(1 - \hat{y}_{fake})] \\ &= -\log(1 - \hat{y}_{fake}) \end{aligned}$$

Therefore:

$$L_{fake} = -\log(1 - D_{\theta}(G_{\phi}(z)))$$

The discriminator wants:

$$D_{\theta}(G_{\phi}(z)) \rightarrow 0$$

Total Discriminator Loss

The real and fake losses are combined:

$$L_D = L_{real} + L_{fake}$$

Therefore:

$$L_D = -\log D_\theta(x) - \log(1 - D_\theta(G_\phi(z)))$$

Equivalently:

$$L_D = -\left[\log D_\theta(x) + \log(1 - D_\theta(G_\phi(z)))\right]$$

Backward Pass: Discriminator

Using the chain rule

Because:

$$L_D = L_{real} + L_{fake} \checkmark$$

the gradient with respect to θ is:

$$\frac{\partial L_D}{\partial \theta} = \frac{\partial L_{real}}{\partial \theta} + \frac{\partial L_{fake}}{\partial \theta} \checkmark$$

$$L_{real} = -\log D_\theta(x)$$

$$L_{fake} = -\log(1 - D_\theta(G_\phi(z)))$$

The gradient flows backward from the total discriminator loss to the discriminator:

$$\underline{L_D \rightarrow D_\theta \rightarrow \theta}$$

The generator is used only to produce the fake image:

$$\hat{x}_{fake} = G_\phi(z)$$

but during discriminator training:

$$\phi \text{ is frozen } \checkmark$$

Therefore, the gradient is not used to update the generator parameters.

Backward Pass: Discriminator

Using the chain rule

Because:

$$L_D = L_{real} + L_{fake}$$

the gradient with respect to θ is:

$$\frac{\partial L_D}{\partial \theta} = \frac{\partial L_{real}}{\partial \theta} + \frac{\partial L_{fake}}{\partial \theta}$$

$$D_\theta \xrightarrow{\hat{y}_{real} = D_\theta(x)} L_{real}$$

Real-image branch

$$\hat{y}_{real} = D_\theta(x)$$

Therefore:

$$\frac{\partial L_{real}}{\partial \theta} = \frac{\partial L_{real}}{\partial \hat{y}_{real}} \cdot \frac{\partial \hat{y}_{real}}{\partial \theta}$$

Substituting:

$$\frac{\partial L_{real}}{\partial \theta} = \frac{\partial L_{real}}{\partial D_\theta(x)} \cdot \frac{\partial D_\theta(x)}{\partial \theta}$$

$$D_\theta \xrightarrow{\hat{y}_{fake} = D_\theta(G_\phi(z))} L_{fake}$$

Fake-image branch

$$\hat{y}_{fake} = D_\theta(G_\phi(z))$$

Therefore:

$$\frac{\partial L_{fake}}{\partial \theta} = \frac{\partial L_{fake}}{\partial \hat{y}_{fake}} \cdot \frac{\partial \hat{y}_{fake}}{\partial \theta}$$

Substituting:

$$\frac{\partial L_{fake}}{\partial \theta} = \frac{\partial L_{fake}}{\partial D_\theta(G_\phi(z))} \cdot \frac{\partial D_\theta(G_\phi(z))}{\partial \theta}$$

Hence:

$$\frac{\partial L_D}{\partial \theta} = \frac{\partial L_{real}}{\partial D_\theta(x)} \cdot \frac{\partial D_\theta(x)}{\partial \theta} + \frac{\partial L_{fake}}{\partial D_\theta(G_\phi(z))} \cdot \frac{\partial D_\theta(G_\phi(z))}{\partial \theta}$$

The discriminator parameters are updated using gradient descent:

$$\theta \leftarrow \theta - \eta \frac{\partial L_D}{\partial \theta}$$

where η is the learning rate.

During this step: θ is updated ϕ is frozen

Numerical example

Suppose:

$$D_{\theta}(x) = 0.9$$

and:

$$D_{\theta}(G_{\phi}(z)) = 0.2$$

Then:

$$L_{real} = -\log(0.9) \approx 0.105$$

$$L_{fake} = -\log(1 - 0.2) = -\log(0.8) \approx 0.223$$

Therefore:

$$L_D = 0.105 + 0.223 = 0.328$$

The discriminator loss is low because the real image is assigned a high probability of being real,

while the generated image is assigned a low probability of being real.

Suppose the discriminator performs poorly:

$$D_{\theta}(x) = 0.2$$

This means that the discriminator assigns only a 20% probability that the real image is real.

Also suppose:

$$D_{\theta}(G_{\phi}(z)) = 0.8$$

This means that the discriminator assigns an 80% probability that the generated image is real.

Loss for the real image

$$L_{real} = -\log D_{\theta}(x)$$

$$L_{real} = -\log(0.2) \approx 1.609 \checkmark$$

The real-image loss is high because the discriminator incorrectly considers the real image to be mostly fake.

Loss for the generated image

$$L_{fake} = -\log(1 - D_{\theta}(G_{\phi}(z)))$$

$$L_{fake} = -\log(1 - 0.8)$$

$$L_{fake} = -\log(0.2) \approx 1.609 \checkmark$$

The fake-image loss is high because the discriminator incorrectly considers the generated image to be mostly real.

Total discriminator loss

$$L_D = L_{real} + L_{fake}$$

$$L_D = 1.609 + 1.609$$

$$L_D \approx \underline{\underline{3.218}}$$

Next Session

GAN Objective and Loss Functions

- GAN Training Procedure
- GAN Convergence and Equilibrium

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Thank you